BPMN MODELING PROJECT · CAMUNDA 8.10

Credit Application: from request to decision

How an automated workflow verifies the customer's file, evaluates the financing project and decides on the credit.

2
parts: customer file verification and project evaluation

3
forms: credit application, project analysis, credit file analysis

14
modeled steps, from start event to customer notification

3
end events: credit granted, credit rejected, application rejected

Objectives and key actors

01 **Standardize**

One traceable path for every request, from submission to customer notification.

02 **Objectify the decision**

Quantified rules: debt-to-income ratio, savings coverage and an overall score out of 100.

03 **Focus human expertise**

The advisor grades the project and only reviews uncertain files (overall score 50 to 70).

POOL · CREDIT APPLICATION

Lane 1 · Customer file verification

- **Customer**: completes the application form
- **Back office**: records and archives documents
- **Camunda engine**: automated checks

Lane 2 · Project evaluation

- **Banking advisor**: project and 50–70 files
- **Camunda engine**: indicators and scores

PART 1 · STEPS 1 TO 6

Verification of the customer’s file

Customer branch

Credit application form

Loan amount, monthly income, savings, loan term (60 months minimum), project description, employment status: all six fields are mandatory.

Bank branch

1. Record the request
2. Receive the documents
3. **check** — Application complete?
4. Archive the documents
5. **check** — Identity, age, banking ban

Immediate rejection

NON CONFORME or NON ELIGIBLE: the customer is notified, end event “Rejection of the application”.

Synchronization (parallel gateway)

Project evaluation starts only when the form AND both checks are complete.

PART 2 · STEPS 7 TO 11

Project evaluation and scoring

7 **Project review**
Advisor: VALIDE/REFUSE, grade 1–10

8 **Financial indicators**
Coverage, monthly payment, debt ratio

9 **Obstructing criterion?**
Debt > 50% or coverage < 10%: reject

10 **Consistency score**
Out of 100, plus a low-debt bonus

11 **Overall score**
70% consistency + 30% project grade

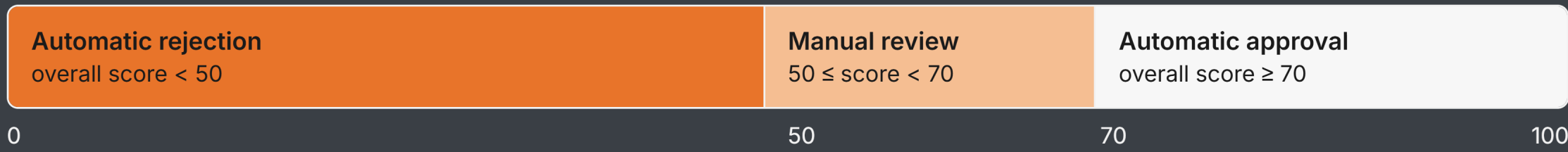
Indicator	Formula in the BPMN
Savings coverage (%)	(Savings ÷ Amount requested) × 100
Monthly payment (€)	Amount requested ÷ Loan term (months)
Debt-to-income ratio (%)	(Monthly payment ÷ Monthly income) × 100
Consistency score (/100)	Debt-ratio points + savings points + low-debt bonus
Overall score (/100)	Consistency × 0.70 + Grade × 10 × 0.30

EXAMPLE

Consistency 80 and project grade 7: (80 × 0.70) + (7 × 10 × 0.30) = **77**, an automatic approval.

GATEWAYS AND THRESHOLDS · STEP 12

Critical decision points



PARALLEL (AND)

Split and join

Customer and bank branches run together; Part 2 waits for both.

EXCLUSIVE (XOR)

Application complete?

CONFORME: continue · NON CONFORME: application rejected

EXCLUSIVE (XOR)

Customer eligible?

Identity, age, banking ban: ELIGIBLE or rejection

EXCLUSIVE (XOR)

Project accepted?

VALIDE: financial calculations · REFUSE: credit rejected

EXCLUSIVE (XOR)

Obstructing criterion?

Debt ratio > 50% OR coverage < 10%: credit rejected

EXCLUSIVE (XOR)

Banking advisor decision

Score 50–70: ACCORD or REFUS, mandatory justification

STEPS 10, 13 AND 14

Scoring grid and process outcomes

Consistency score (max. 100 points)

Debt-to-income ratio	Points	Savings coverage	Points
≤ 30%	50	≥ 75%	40
> 30% to 35%	40	≥ 50% to < 75%	30
> 35% to 40%	25	≥ 25% to < 50%	20
> 40% to 50%	10	≥ 10% to < 25%	10
> 50%	0	< 10%	0

+10 bonus points when the debt-to-income ratio is at or below 10%.

Three possible end events

Credit granted

Score ≥ 70 or advisor ACCORD, then agreement notification

Credit rejected

Project REFUSE, obstructing criterion, score < 50 or advisor REFUS

Application rejected

Incomplete file or ineligible customer in Part 1

Out of scope: contract signing and fund disbursement.